

Appendix 1: FluorMod tutorial

This appendix presents a brief tutorial on using the Regress-Then-Sum method in FluorMod (v4.0) to solve the linear mixing problem with EEMs. It should be noted that the Regress-Then-Sum (RTS) classification scheme is not specific to fluorescence, though that is the problem addressed for this tutorial. Hence, the RTS approach can be applied to any spectral information. The tutorial is presented for users to see how EEMs were vectorized, how to model EEMs with RTS using a library of known sources, and then examine the output of results and model fit diagnostics. The code in this tutorial was written in the R program (version 4.5.1.) and RStudio (2025.05.01) as the computing environment. Google Gemini Pro (license through NC State University) was used to improve speed, accuracy, assess model performance, and create a coherent workflow from original R scripts. The R code contained in this tutorial is free software that can be modified to the user's needs under the GNU General Public License, with appropriate citation to this publication. For example, modifications to the range of excitation and emission wavelengths and spectral resolution may be required when they do not match the instrument settings described in the next section.

Ten (10) EEMs used in the main text are made available as unknown samples for this tutorial (Table A1).

Table A1. List of EEMs from the main text used in this tutorial. Locations and dates are included to cross reference with the data. 'eem' is the order in the input data array, 'filename' is the CSV file name, 'stn' is the station (NR0 or NR160) sampled in the Neuse River Estuary, and 'date' is the sampling date. Results obtained in this tutorial thus may be compared to Table S6.			
<u>eem</u>	<u>filename</u>	<u>stn</u>	<u>Date</u>
1	p_01.csv	NR0	4/12/2010
2	p_02.csv	NR160	4/12/2010
3	p_03.csv	NR0	4/9/2012
4	p_04.csv	NR160	4/9/2012
5	p_05.csv	NR0	4/6/2016
6	p_06.csv	NR160	4/6/2016
7	p_07.csv	NR0	4/19/2016
8	p_08.csv	NR160	4/19/2016
9	p_09.csv	NR160	4/20/2017
10	p_10.csv	NR160	4/11/2018

Preprocessing of EEMs. The data to be vectorized and modeled are a subset of 10 EEMs that have been corrected for instrument biases (excitation lamp intensity and emission detector response) and inner filter effects, have had a corresponding ultrapure laboratory grade water blank (Milli-Q) subtracted, and calibrated to the instrument's water Raman signal and expressed in quinine sulfate units (QSU; see methods in the main text). Rayleigh and Raman scattering signals have been replaced with NaN values (Stedmon and Bro 2008a). The excitation wavelength range (240 to 450 nm in 5 nm increments) and emission wavelength range (300 to 600 nm in 2 nm increments) are the same as in the main text.

Note on p_files. The aim of this tutorial is to assist the user with loading processed EEMs (herein referred to as 'p-files' per the convention used in the authors' workflow). The p_file is a CSV file format containing only corrected fluorescence intensity values and NaN values that R reads as text and then converts to appropriate missing values. There are no headers or other information (e.g., excitation or

emission wavelengths) in these p_files. The user is encouraged to open and view these files to understand their format prior to working with them. Users may need to adapt FluorMod to their own workflow (e.g., change the code to search for their versions of p_files).

In this tutorial, commands and conventions were written on a PC running Windows 10. The R code used in RStudio is written in blue with [Lucida Console font](#). It is assumed that the user has some basic understanding of the R programming language, but these instructions are presented so the novice can execute them line by line in a working script or simply enter each command in the R console. For assistance with Rstudio, see the RStudio User Guide (<https://docs.posit.co/ide/user/>).

Tutorial. You will need an internet connection to access the tutorial and to install certain R packages onto your computer. Once this is done, the tutorial can be completed offline.

- A. Download the FluorMod-v4.0.zip file (<https://doi.org/10.5281/zenodo.18792488>) to your working directory. You will need to extract the folder, which will be named “closburn-FluorMod-7aff40”. Rename it “fm_tutorial”. Start RStudio (or R) and set your working directory to the location of fm_tutorial. For example:

```
setwd("C:/Users/closburn/Documents/Projects/FluorMod-v4.0/fm_tutorial")
```

- B. The fm_tutorial folder has the following folder structure, which should be replicated for future use of FluorMod:

```
/fm_tutorial
|-- /data
|   |-- /raw_eems # Put your sample .csv files here (no headers needed)
|   |-- source_eems.csv # Your library of source EEMs (columns = sources)
|-- /scripts
|   |-- reshaper_tool.R # Helper 1: Vectorizes data
|   |-- FluorMod_RTS.R # Helper 2: Runs the NNLS model
|-- .gitignore # Github file you may ignore.
|-- FluorMod.Rproj #RStudio project file
|-- fm_workflow.R # MASTER SCRIPT (Run this one!)
```

A “results” folder is not present but will be created later in the fm_tutorial directory.

NOTE: this tutorial already contains a source_eems.csv file that has been vectorized. Users will need to modify reshaper_tool.R with `output_file = “source_eems.csv”` and run the script manually. Then, in the header (top) row of their own “source_eems.csv” file, change the filename to the specific source category, e.g., “Wetland”. The source_eems.csv file provided for this tutorial can be viewed to see how to set up a new source file.

- C. Before we go line by line in the script “fm_workflow.R”, in practice FluorMod can be run by entering the following command at the RStudio console prompt:

```
source("fm_workflow.R")
```

Try this now and you should see this result:

```
> source("fm_workflow.R")
--- Step 1: Loading Data ---
Found 10 files. Processing...
```

Success! Saved as: unknown_eems.csv Rows: 6493 Columns: 152

Column specification

Delimiter: ","

dbl (152): Soil, River, Soil, Septic, Septic, Septic, Septic, Septic, Septic, Septic, Sep...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

--- Step 2: Running FluorMod RTS ----- Analysis Complete! ---[1] "Top 5 Samples by Fit (R2):"

eem_name	R2	RMSE
p_02	0.9966106	0.4716436
p_04	0.9965441	0.7960817
p_03	0.9941159	0.3063562
p_10	0.9918382	0.7828070
p_01	0.9838684	1.0655978

FluorMod shows two performance metrics in the table, the coefficient of variation (R2) and root mean square error (RMSE).

- D. Open "fm_workflow.R". You will see a header "FM_WORKFLOW: Master Script for FluorMod Analysis". The preceding '#' symbols flag this text as comments for informational purposes and R ignores them.
- E. Step #1 is SETUP. Highlight lines 10 to 15 in the script and select Code – Run Selected Line(s), or press CTRL+O.

R checks if the 'nnls' and 'readr' packages are installed and then installs and attaches them if not installed.

```
if (!require("nnls")) install.packages("nnls")
if (!require("readr")) install.packages("readr")
library(nnls)
library(readr)
```

R loads the helper scripts into the working Environment.

```
source("scripts/reshaper_tool.R")
source("scripts/FluorMod_RTS.R")
```

- F. Step #2 is LOAD DATA. Highlight and run lines 21 to 33.
 - 1. Vectorize the unknown samples, assuming raw CSV files are in "data/raw_eems".

```
unknown_data <- reshape_eems(
  input_path = "data/raw_eems",
  file_pattern = "^p_.*\\.csv$",
  output_file = "unknown_eems.csv")
```

You can change 'file_pattern' to "*.csv\$"; the current flag looks specifically for p-files. An output file "unknown_eems.csv" is written into the current directory.

You should see 'unknown_data' in your Environment and "unknown_eems.csv" in your fm_tutorial folder. You should also see this output in the Console: **Found 10 files. Processing... Success! Saved as: unknown_eems.csv**

2. Load the source library.

```
source_data <- readr::read_csv("data/source_eems.csv",  
                               name_repair = "minimal")
```

You should see the following output in the Console:

```
Rows: 6493 Columns: 152  
— Column specification —————  
Delimiter: ","  
dbl (152): Soil, River, Soil, Septic, Septic, Septic, Septic, Septic,  
           , Septic, Septic, Sep...  
  
i Use `spec()` to retrieve the full column specification for this data.  
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

There are ten sources in source_eems.csv: Soil, River, Septic, Effluent, Wetland, Influent, Street, Swine, and Poultry. As discussed in the main text, Soil, River, and Wetland comprised most of the sources for the majority of samples, though there were some exceptions.

G. Step #3 is RUN MODEL. Highlight and run lines 37 – 53 to apply the RTS model to the 10 vectorized EEMs.

1. SAFETY CHECK: Create a 'results' folder if it doesn't exist.

```
if (!dir.exists("results")) {  
  dir.create("results")  
  message("Created 'results' directory.")  
}
```

2. Create a run identified (ID) so results will save to the appropriate "results/" folder.

```
run_id <- paste0("results/FM-", format(Sys.Date(), "%Y%m%d"))
```

3. Run the FluorMod_RTS function. The inputs are the source_eem_data, the unknown_eem_data, and the run_id which is the prefix of the output file.

```
results <- FluorMod_RTS(  
  source_eem_data = source_data,  
  unknown_eem_data = unknown_data,  
  output_prefix = run_id
```

)

The output_prefix will be "FM_YYYYMMDD_*.csv".

H. Step #4 is REVIEW. Highlight and run lines 56 – 58 to review the results.

```
message("--- Analysis Complete! ---")
print("Top 5 Samples by Fit (R2):")
print(head(results$pcts[order(-results$pcts$R2), c("eem_name", "R2",
"RMSE")], 5))
```

You should see this output in the Console:

```
> print("Top 5 Samples by Fit (R2):")
[1] "Top 5 Samples by Fit (R2):"
> print(head(results$pcts[order(-results$pcts$R2), c("eem_name",
"R2", "RMSE")], 5))
```

```
      eem_name      R2      RMSE
p_02      p_02 0.9966106 0.4716436
p_04      p_04 0.9965441 0.7960817
p_03      p_03 0.9941159 0.3063562
p_10      p_10 0.9918382 0.7828070
p_01      p_01 0.9838684 1.0655978
```

The output is the same as the output listed above when FluorMod was run in its entirety in Step C.

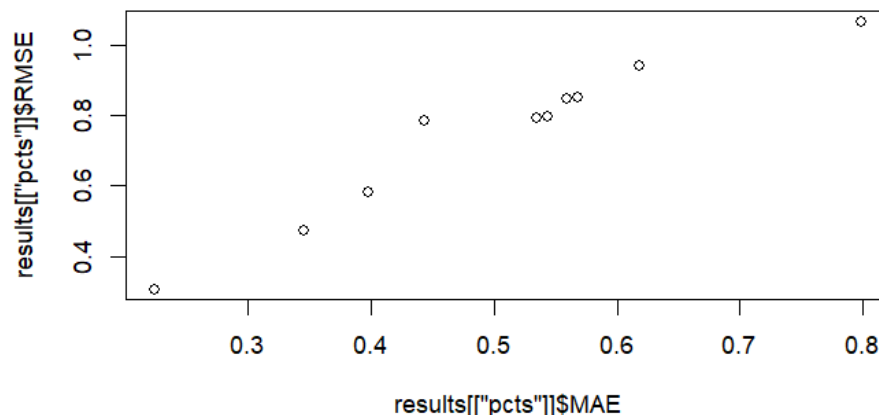
- Three output files are generated. Here is a description of each file.
 - FM_YYYYMMDD_individual_weights.csv – This file contains each regression coefficient for each entry in the dictionary. There will be as many columns as there are entries in the source_eems.csv file.
 - FM_YYYYMMDD_summed_weights.csv – This file contains the summed regression coefficients across categories.
 - FM_YYYYMMDD_pcts.csv – This file contains the normalized weights expressed as a fraction between 0 and 1 (or 0 and 100%).
- Most users will work with the 'pcts' file so open that file in Excel if you wish. Otherwise, type this command in the RStudio Console to view it: `view(results[["pcts"]])`. You should see the following:

	eem_name	Soil	River	Septic	Effluent	Wetland	Influent	Street	Swine	Poultry	R2	Fmax	RMSE	MAE
p_01	p_01	0.01880180	0.000000000	0.000000000	0.000000000	0.89491489	0.086283303	0.000000000	0.000000000	0.000000000	0.9838684	38.14130	1.0655978	0.7982728
p_02	p_02	0.04220333	0.742200884	0.000000000	0.000000000	0.01264397	0.047408920	0.15554290	0.000000000	0.000000000	0.9966106	42.24138	0.4716436	0.3456698
p_03	p_03	0.04597777	0.515384029	0.0186986566	0.000000000	0.01445965	0.017067714	0.37417860	0.008510648	0.005722933	0.9941159	24.01224	0.3063562	0.2234643
p_04	p_04	0.04847604	0.762189121	0.000000000	0.000000000	0.14628827	0.001188923	0.04185764	0.000000000	0.000000000	0.9965441	82.17030	0.7960817	0.5440036
p_05	p_05	0.18000033	0.003961785	0.000000000	0.000000000	0.81603788	0.000000000	0.000000000	0.000000000	0.000000000	0.9806495	39.99289	0.9405445	0.6179927
p_06	p_06	0.16861366	0.023835458	0.0002956099	0.000000000	0.79603673	0.000000000	0.000000000	0.000000000	0.011218540	0.9781371	33.48298	0.8476655	0.5592021
p_07	p_07	0.12320037	0.010683679	0.000000000	0.000000000	0.86611596	0.000000000	0.000000000	0.000000000	0.000000000	0.9810956	36.78003	0.8497475	0.5684808
p_08	p_08	0.16075115	0.036474651	0.0009139631	0.000000000	0.76768801	0.001046947	0.02762074	0.000000000	0.005504536	0.9798333	33.15269	0.7914738	0.5350709
p_09	p_09	0.13809227	0.034181081	0.000000000	0.000000000	0.71563562	0.016492729	0.07811286	0.000000000	0.017485438	0.9759668	21.78222	0.5809164	0.3977199
p_10	p_10	0.12484491	0.026105366	0.2552388823	0.03081187	0.49183252	0.000000000	0.07116645	0.000000000	0.000000000	0.9918382	50.78586	0.7828070	0.4433813

- i. Columns 2-10 are the normalized weights (from 0 to 1) for each sample. For example, sample p_01 (NR0 from 4/12/2010) has Soil = 0.019, Wetland = 0.895, and Influent = 0.086.
- ii. R2 (i.e., R^2) is the coefficient of variation; we can think of this as the "shape" metric, because it measures how well the model replicated the unique fingerprint (or shape) of the EEMs. Results range from 0.976 to 0.997, which are very high scores and indicate that the source_eem dictionary accurately represents the sources comprising the unknown samples. Scores such as 0.997 represent near perfect match between the predicted EEM and the actual EEM.
- iii. Fmax is the maximum fluorescence signal intensity (in QSU) in an unknown sample.
- iv. RMSE is the root mean square error and in FluorMod it can be thought of as the "peak intensity" metric because it measures the average error in intensity units (QSU). Large errors are penalized more heavily; i.e., if the model was missing a strong signal, RMSE will spike. Results in this example ranged from 0.306 to 1.066. Sample p_03 (RMSE = 0.306) had the best fit in terms of fluorescence intensity. FluorMod almost perfectly predicted the height of every peak in the EEM. In contrast, sample p_01 (RMSE = 1.066) had the highest error. Though the R^2 value (0.984) indicates the FluorMod captured the shape correctly, it did less well capturing the intensity. This often happens if there is a concentration effect (like Inner Filter Effect) or if the sample is simply much brighter than the others, making the residuals naturally larger.
- v. MAE is the mean absolute error - the average difference between the real unknown sample and FluorMod's prediction. It is less sensitive to outliers than RMSE. Values ranged from 0.22 to 0.80 and track well with the RMSE values, which you can observe by typing:

```
plot(results[["pcts"]] $\$$ MAE, results[["pcts"]] $\$$ RMSE).
```

You should see the following:



This result indicates there are no strange outliers and that the error is consistent across the samples.

- vi. Let's consider sample p_01 again. FluorMod used the source_eem dictionary to represent the sources well, but it did struggle a bit with how much fluorescence was present. This sample might contain a unique component not fully represented in the dictionary (source_eem file). However, because the R^2 is still 0.98, we can still have confidence in the % composition results.

We might wish to explore the limitations of FluorMod's results in terms of the relative error for each sample. One way to do this is to use a limit of detection (LOD) metric, common in spectroscopy. A typical convention is that the LOD is three times the noise. In this case, $LOD = 3 \times RMSE$. The LOD divided by the Fmax value gives a rough percentage cutoff for valid contributions. This calculation can be done by the user from the results in the FM_YYYYMMDD_pcts.csv file.